

The "Lucky-Fold" Trap in EEG Classification: Why Multi-Resample LPSO (P=6) + Median±IQR Gives Honest Cross-Subject Performance

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Abstract. Reported EEG classification accuracies often exceed 90% under within-subject 80/20 splits, but such evaluations can be dominated by identity leakage, where models learn person-specific signatures (fingerprints) and then appear to classify disease. We treat within-subject scores as a control for fingerprintability and treat cross-subject generalization as the clinical target. We use Leave-P-Subjects-Out (LPSO) at P=6 with many resamples (Random-50) and summarize performance using median $\pm$ IQR across folds. On 65 subjects from OpenNeuro ds004504 (Alzheimer's disease vs Control), we compare two feature pipelines under identical LPSO: PCA over standardized PSD features versus univariate F-test selection over channel-by-band power features. Under Random-50 LPSO, F-test improves median accuracy over PCA by **___ points** [Δ = ___, 95% CI ___ to ___, p = ___, Cliff's δ = ___]. Single LPSO folds range from **39.8%** to **98.3%** on the same model, illustrating a lucky-fold illusion that a single split can conceal. We recommend multi-resample subject-disjoint evaluation, subject-level metrics, robust summaries (median $\pm$ IQR), and public split manifests as default reporting practice for EEG studies that claim deployable performance.

1. Narrative overview

Clinical question. In practice, models are deployed on new people. The relevant question is not whether a model can separate epochs drawn from the same participant, but whether it can generalize to participants never seen during training.

Core claim. Many near-ceiling EEG accuracies arise from evaluation choices rather than model advances. When train and test sets share participants, models can exploit stable identity signatures and condition prediction can look deceptively strong.

What we test. If identity leakage is a major driver, we expect: (i) within-subject 80/20 scores near-ceiling for both fingerprinting and condition; (ii) subject-disjoint LPSO medians drop to an honest range; (iii) single LPSO folds are unstable; (iv) multi-resample summaries stabilize estimates; and (v) biologically grounded channel-by-band features generalize better than variance-only transforms.

2. Related work (short)

We ground the evaluation argument in work documenting leakage from segment-level splitting and subject overlap, and work showing that data partitioning and cross-validation settings can substantially change reported EEG model performance. We also cite EEG biometrics literature showing identity can be predicted reliably from EEG, supporting fingerprinting as a diagnostic control.

3. Data, preprocessing, and subject-level metric

Dataset. OpenNeuro ds004504; 65 subjects used (36 AD, 29 Control). Eyes-closed resting EEG.

Signal pipeline. Welch PSD; 3.0 s windows; 0.5 s overlap; PSD normalization; artifact rejection (reject = 600 μ V, flat = 0.3). Bands: Delta 0.5-4 Hz, Theta 4-8 Hz, Alpha 8-12 Hz, Beta 12-30 Hz.

Subject-level decision. Epoch predictions are aggregated per subject using an AD-ratio threshold selected from a fixed grid. (Specify: global threshold vs fold-specific thresholding and how leakage is prevented.)

4. Feature pipelines

PCA. z-score then PCA retaining ____% variance, then train model. PCA captures major variance directions, but these may reflect nuisance structure or identity rather than disease signal.

Univariate F-test. One-way ANOVA F-test per channel-by-band feature (two classes implies proportionality to t^2). Apply family-wise control at $p < 0.05$ (specify method) then MinMax scale. This preserves interpretable channel-by-band features that already separate groups.

Retention placeholders. Mean features retained = ____ of 95; median = ____; IQR = ____.

5. Evaluation protocol

5.1 Controls: within-subject 80/20

These controls are not deployable evidence. They quantify how easily identity can be exploited.

Fingerprinting (predict identity). Near-ceiling: ____% [95% CI ____ to ____.]

Condition within same person. Mirrors fingerprinting: ____% [Δ vs fingerprinting = ____, p = ____.]

Figure 1 placeholder: within-subject 80/20 fingerprinting (PCA vs F-test).

Figure 2 placeholder: within-subject 80/20 condition prediction (PCA vs F-test).

5.2 Cross-subject LPSO at P=6

Strategies. Systematic-12 (deterministic blocks) versus Random-50 (independent resamples). Random-50 estimates the distribution over possible subject partitions and exposes tail risk.

Same-subjects control. When comparing strategies, filter to comparable test-subject sets where needed (details: ____).

Figure 3 placeholder: small-multiples boxplots (rows: PCA vs F-test; cols: Systematic-12 vs Random-50) with fold dots and chance line.

Primary comparison (Random-50, LPSO P=6). Median accuracy: F-test = ____% [95% CI ____ to ____], PCA = ____% [95% CI ____ to ____]. Difference Δ = ____ points [p = ____, Cliff's δ = ____].

Dispersion: IQR(F-test) = ___, IQR(PCA) = ___; min/max(F-test) = ___/___, min/max(PCA) = ___/___.

Single-fold instability (lucky folds). Worst fold = 39.8% (held-out: sub-___, sub-___). Best fold = 98.3% (held-out: sub-___, sub-___). A single reported fold can therefore be unrepresentative on this cohort size.

Figure 4 placeholder: histogram or density of fold accuracies (Random-50, P=6) showing heavy tails and lucky-fold region.

Table 1 placeholder: LPSO P=6 medians $\pm$ IQR and min/max per model (MLP, XGBoost, SVM, KNN) for PCA vs F-test, Systematic-12 vs Random-50.

5.3 Convergence with number of folds and resamples

A single fold is one draw from a broad outcome distribution. As the number of resamples increases, the median stabilizes and uncertainty shrinks. Even 2 to 3 folds are materially more informative than one, while 10 or more provide a more reliable central tendency for moderate cohorts.

Figure 5 placeholder: running median accuracy vs number of resamples k ($k = 1, 2, 3, 5, 10, 20, 50$) with running IQR or bootstrap CI band.

5.4 Effect of hold-out size P

Compare P=6 and P=2 under Random-50. Report medians, IQR, and extremes. Working expectation: medians may be similar, but variance and extremes inflate at P=2 due to smaller test sets per fold.

Table 2 placeholder: variance vs P (Random-50), rows: F-test P=6, F-test P=2, PCA P=6, PCA P=2.

6. Biomarkers (subject-level)

Summarize top channel-by-band differences at the subject level. Working hypothesis: posterior alpha reductions and broader slowing signatures appear in AD cohorts, consistent with prior literature. Provide effect sizes, confidence intervals, and corrected p-values where possible.

Figure 6 placeholder: posterior alpha biomarkers (box or violin + swarm) with group means and 95% CI whiskers.

Table 3 placeholder: top 5 to 8 biomarkers with AD/control means, percent delta, Cohen d , 95% CI, p-value (corrected).

7. Discussion

What the controls show. Fingerprinting is near-ceiling under within-subject splits, and within-subject condition prediction tracks fingerprinting. This supports the claim that identity overlap can explain inflated within-subject disease results.

What LPSO shows. When identities are disjoint, medians drop to an honest range and the ranking of methods changes; dispersion is large. This is the performance regime that aligns with deployment.

Why Random-50. Tail risk (39.8% to 98.3%) is large; single splits are not representative. Median $\pm$ IQR over many resamples is robust, easy to read, and hard to game.

Field recommendation. For EEG studies claiming deployable generalization, report subject-disjoint evaluation with multiple folds or resamples, include robust summaries and uncertainty, and publish split manifests.

8. Threats to validity

Single cohort and paradigm. Eyes-closed resting; results may differ for task EEG, other pathologies, or different sensors.

Thresholding. Subject decision uses AD-ratio thresholding; operating points may shift by cohort. Specify how thresholding avoids train-test leakage.

Confounds. Age, medication, and recording noise are potential confounders. Covariate regression was not done here (plan: ____).

9. Reproducibility

Release JSON split manifests for Random-50 at $P=6$ and $P=2$, plus scripts to regenerate features and figures. Use containerized environments and versioned configs. Provide dataset hash, code commit, container digest, and seed policy.

10. Conclusion

Within-subject near-ceiling scores are consistent with identity overlap effects. Under subject-disjoint LPSO, medians fall to honest ranges; univariate F-test over bandpower features outperforms PCA by ____ points [95% CI ____ to ____, $p =$ ____, $\delta =$ ____]. Report median $\pm$ IQR over many resamples, publish split manifests, and treat within-subject 80/20 as a fingerprintability control rather than deployable evidence.

References (starter set to cite in BibTeX)

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